Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

MACHINE LEARNING

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory and carry marks as indicated.

(2) Assume suitable data wherever necessary.

(3) Illustrate your answers with neat sketches.

(4) Use of Non – programmable calculator is permitted.

1. (a) The prior probabilities are to be estimated from this randomly sampled data :

Class	A	A	B	A	B
x	1	2	1	2	1
y	2	2	2	1	1

(i) Apply Bayes' Theorem to estimate the probability that a sample with $x = 1$, $y = 2$, came from class A.

(ii) Assume x and y are independent and then estimate the probability that a sample with $x = 1$, $y = 2$, came from class A. 4(CO2,4)

(b) Describe the components of a Face Recognition system. 6(CO1,2)

2. (a) Write a Python script to read a csv file named "sensors.csv" and convert the read data into data frame. Print the different statistical parameters like mean, variance, covariance, correlation matrix of the loaded data set. 4(CO3,4)

(b) Write a Python script to read a csv file named "heart_rate.csv" and convert the read data into data frame. Look for categorical features and encode them using label encoding. Visualize features using pairplot. Apply scaling to all features and apply log transform to unskew the features for skewness above 0·8. 6(CO3,4)

3. (a) Write a Python script to split the data present in X (features) and Y (targets) into 65% train and 35% test set. Use cross validation score to train a generalized classifier using Support Vector Machine, Quadratic Discriminant Analysis and Linear Discriminant Analysis techniques. Include the statement to plot the confusion matrix and print the sensitivity and specificity values. 6(CO3,4)

(b) Write a Python script to predict the house sale price using "housing_price.csv". Apply linear regression and SVM regressor with cross validation and print the error metrics. 4(CO3,4)

4. (a) How would a sample with a feature vector (2, 1) be classified if samples from class A are at (3, 3), (4, 2) and (3, 3) and samples from class B are at (2, –1) and (3, –1·5) and samples from class C are at (2, 1), (3, 2) :

(i) Using KNN (k = 1) and Euclidean distance.

(ii) Using KNN (k = 2) and city block distance.

Compare the results and comment on the results. 6(CO4)

(b) Estimate a density function using a symmetric triangular kernel with a base width of 2, given that your samples are at 2, 3, 3 and 4. 4(CO2)

5. (a) Perform a hierarchical clustering of the five samples using the single linkage algorithm :

	1	2	3	4	5
x	4	24	15	4	24
y	8	12	8	8	12

Show the values of the centroids and the distances from the samples to the centroids. 7(CO2,4)

(b) Explain two input OR-networking using example. 3(CO2)

6. (a) As a machine learning consultant, you have been approached by defense contractor to design an automated robot to detect and defuse bombs. Explain the different components required for the system with block diagram. Which features and how many will you choose ? What are your challenges and how will you overcome them ? 7(CO1,5)

(b) Differentiate between Machine Learning and Deep Learning using example. 3(CO2)